Final Report

Electricity Price Trends and Market Volatility in Europe

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1 Executive Summary

1.1 Introduction

In recent years, European electricity markets have become increasingly volatile and structurally complex. Rising shares of renewable generation, fluctuating fuel prices, and shifts in cross-border flows have made price patterns harder to interpret and predict (International Energy Agency, 2023). For market participants and analytics companies like MARBL, who support flexibility providers and traders with data-driven insights, accurately understanding and anticipating these dynamics is essential for optimizing trading performance and mitigating financial risks.

To operate effectively in this environment, MARBL requires both a structural perspective on market behaviour and robust predictive tools. Insights derived from recurring price shapes help reveal fundamental daily dynamics, while short-term forecasting models translate these insights into actionable expectations about future price movements. Together, they form a coherent link between market interpretation and predictive analytics.

The objective of this project is to detect repeating structures in day-ahead price profiles and use them as the basis for developing forecasting models in selected European wholesale markets, namely Norway, Denmark, and Spain. Using hourly day-ahead price data obtained from the European Network of Transmission System Operators for Electricity (ENTSO-E) Transparency Platform, daily price curves will be converted into 24-hour profiles and grouped using hierarchical clustering to identify characteristic market patterns. These clusters will then be combined with weather-related features to build XGBoost-based day-ahead forecasting models, generating both structural insights and predictive outputs for the three markets. By following established methodologies from prior research, the project expands MARBL's analytical scope and strengthens its capabilities in market intelligence and data-driven trading support.

1.2 Project Scope

This section outlines the exact boundaries of the project by specifying what the work will cover and what falls outside its remit.

1.2.1 In Scope

The primary objective of this project is to generate structural insights from day-ahead electricity price profiles and to leverage these insights for short-term price forecasting in selected European electricity markets. The scope therefore explicitly encompasses both the identification of recurring price structures and their use within a predictive framework.

To address this objective, the project focuses on the analysis of temporal and structural characteristics of day-ahead electricity prices in Norway, Denmark, and Spain. The analysis considers recurring daily price patterns derived from hourly day-ahead price data and uses these patterns to support market interpretation and short-term price prediction. The structural analysis is conducted over a three-year observation period. Although a longer historical horizon was initially considered, the timeframe was deliberately restricted to the most recent three years in order to limit the influence of structural market changes and non-stationary dynamics, thereby ensuring more consistent and interpretable results. Within this scope, the project includes the derivation of daily price pattern groups and the generation of visual representations that support the interpretation of market behaviour.

Based on these structural insights, the project proceeds with the development of forecasting models using established machine learning techniques, with XGBoost as our primary algorithm. Additionally, the scope included building a real-time data pipeline capable of feeding a daily visualization dashboard, enabling continuous monitoring of updated price information. After consultation of the project partner, it was mutually agreed to feed the visualization dashboard with the insights generated over the course of the project, but not necessarily with real-time price predictions. Instead, it was agreed to formulate recommendations for next steps for a possible follow-up project.

1.2.2 Out of Scope

To keep the project within the planned timeframe, several elements are intentionally excluded. The analysis will not extend to other market layers such as balancing or retail markets. Likewise, the forecasting models will not incorporate certain external explanatory features, including fuel prices or CO_2 price data. The project will also refrain from providing causal interpretations of market drivers, as the emphasis lies on predictive performance rather than econometric justification. The full-scale technical deployment and integration into MARBL's production environment is beyond the scope.

1.3 Milestones

To structure the project workflow and ensure steady progress, a set of clearly defined milestones was established. With the exception of minor delays arising during the data integration and preprocessing phase, the project largely adhered to the planned timeline, and the defined milestones were met as intended.

Milestone	Date	Definition	Status
M1	26.10.2025	First Project Plan	Done
M2	7.11.2025	Final Project Plan	Done
M3	12.11.2025	Data Integration & Pre-processing	Done
M4	19.11.2025	Cluster Analysis	Done
M5	26.11.2025	Intermediate Report	Done
M6	8.12.2025	Trained Models	Done
M7	22.12.2025	Mix-of-Experts Model	Done
M8	5.1.2026	Day-Ahead-Price Prediction Model	Done
M9	15.01.2025	Visualization board	Done
M10	23.1.2025	First Project Report & Slides	Done
M11	2.2.2026	Final Project Report	Done

The full source code, covering the entire project lifecycle—including data ingestion pipelines, cluster analysis, prediction modeling, and the Streamlit dashboard—is available in the project repository: https://github.com/builtbymaxim/DsLab25W_marbl.energy

2 Data Insights

The dataset comprised approximately 262,800 hourly day-ahead electricity price observations (3,650 days × 24 hours × 3 bidding zones) covering a ten-year period across the selected markets. After preprocessing, this resulted in roughly 10,950 daily price profiles (3,650 per zone), which were used as input for the clustering analysis. The size of this dataset provided sufficient statistical power for robust model training, cross-validation, and comprehensive evaluation of the applied methodology.

The selected bidding zones represent structurally diverse European electricity markets, enabling an assessment of the generalizability of the clustering approach. Spain (ES) represents a market with high solar penetration, where solar irradiance exhibits a statistically significant negative relationship with electricity prices. Norway (NO2) is characterized by a generation mix dominated by hydropower, which accounted for approximately 88% of total electricity generation in 2022. Denmark (DK1) represents a system with exceptionally high wind penetration, with wind energy contributing around 55% of electricity generation in 2022 (Mosquera-López et al., 2024).

The ten-year observation period captured multiple seasonal cycles, structural market developments such as renewable energy expansion and coal phase-outs, varying fuel price regimes, and several extreme weather events. This temporal coverage exceeds the commonly recommended minimum of one to two years for reliable seasonal pattern identification (Roberts & Brown, 2020).

With respect to data quality, all selected bidding zones (ES, DK1, NO2) belong to established and highly liquid European power markets, including IBEREL and Nord Pool, and provide reliable price reporting via the ENTSO-E Transparency Platform. Observed data gaps and outliers were systematically identified and treated during the preprocessing stage. In addition, the ERA5 reanalysis dataset offered complete temporal coverage and consistent meteorological data quality. Its reliability is ensured by the European Centre for Medium-Range Weather Forecasts (ECMWF) and has been extensively validated against ground-based observations (Hersbach et al., 2020), making it a state-of-the-art source for weather-related features. For the live forecasting component, WeatherAPI.com was used as a reliable and accessible provider of short-term meteorological forecast data.

3 Licenses, and legal and ethical concerns

Day-ahead electricity price data from the ENTSO-E Transparency Platform is provided under Creative Commons Attribution 4.0 International License (CC-BY 4.0), permitting free copying, redistribution, and commercial use with required attribution: "Data source: ENTSO-E Transparency Platform" (ENTSO-E Transparency Platform, 2023). ERA5 climate reanalysis data from the Copernicus Climate Change Service is released under Copernicus License, permitting free commercial use with required attribution: "Contains modified Copernicus Climate Change Service information [Year]" and disclaimer that neither the European Commission nor ECMWF is responsible for use of the data (ECMWF, 2024). The WeatherAPI.com forecast data is available under a free tier for development purposes, subject to the provider's standard Terms of Service (WeatherAPI.com, 2025).

Only aggregated market and meteorological reanalysis data are processed—no personal data or commercially sensitive information is involved, therefore GDPR does not apply. All methods use publicly available scientific literature and open-source software and packages without proprietary algorithms. Only public information from official channels (ENTSO-E, Copernicus and WeatherAPI.com) is used, with no insider information or privileged access. Results support MARBL's market intelligence within regulatory frameworks, such as MAR (Market Abuse Regulation) and REMIT (Regulation on Wholesale Energy Market Integrity and Transparency) by enhancing understanding of public patterns rather than exploiting information asymmetries.

4 Solution Architecture

The solution architecture is structured into three core components: data integration and preprocessing, pattern detection, and predictive modelling. Figure 1 provides a high-level overview of the overall architecture, illustrating the pattern detection and predictive modelling.

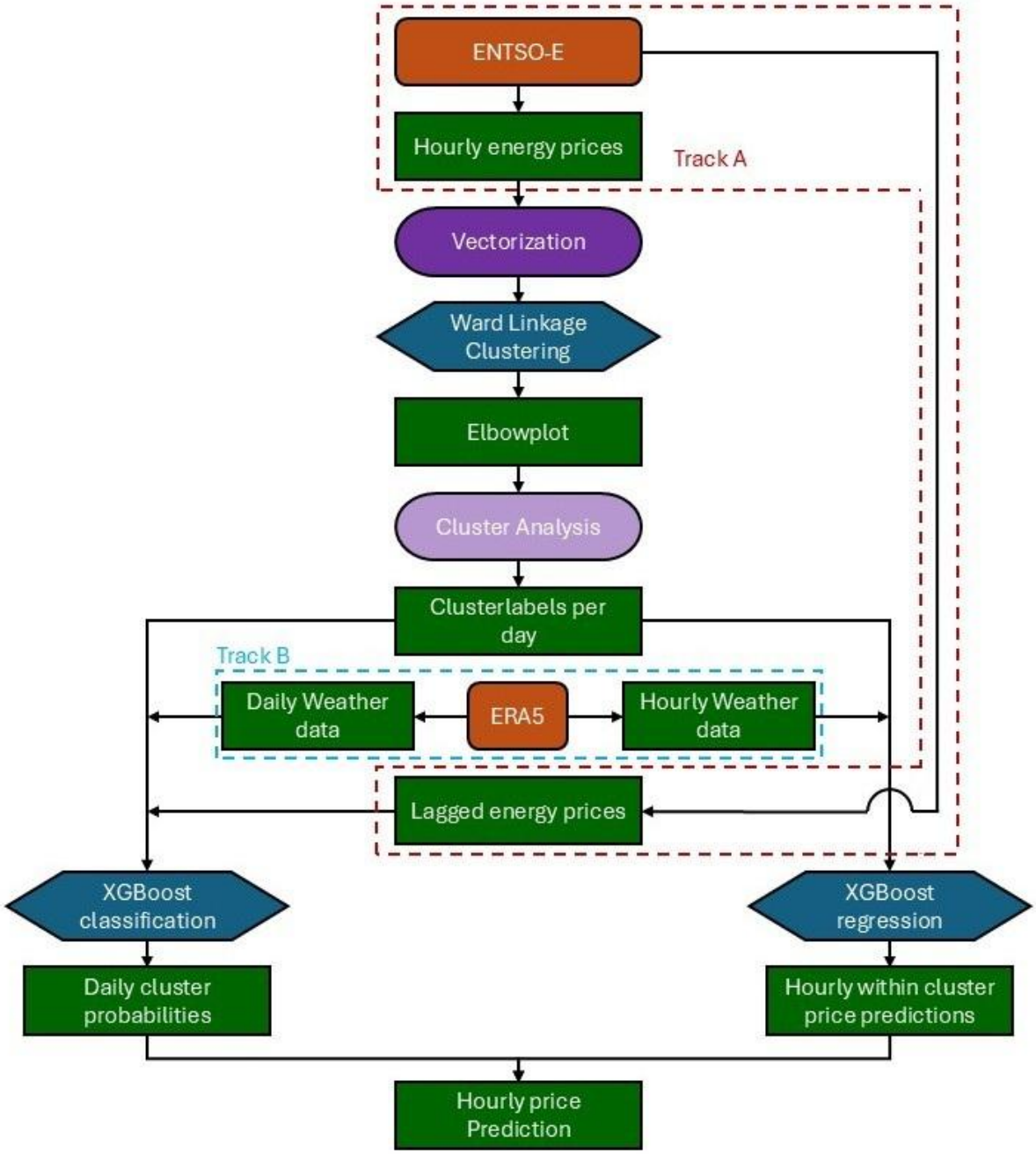

Figure 1 A simplified visualization of the solution architecture.

4.1 Data Integration and Preprocessing

To ensure the system is reliable and flexible, the data architecture was designed as a Parallel ETL (Extract, Transform, Load) Pipeline. The system processes market prices (Track A) and grid-based weather data (Track B) independently. This separation prevents a failure in one data source from stopping the other, with data converging only at the final merging stage.

Three distinct bidding zones were selected to represent different types of power generation: Denmark West (DK1) for wind power, Spain (ES) for solar power, and Southern Norway (NO2) for hydro storage.

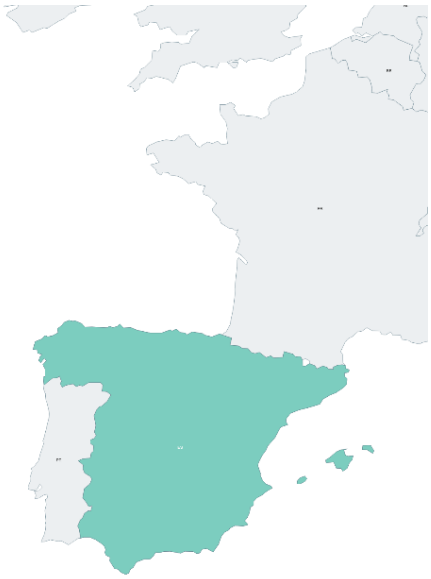

Figure 3 Bidding Zone: ES

Figure 2 Bidding Zones: DK1 (green), NO2 (blue), NO4 (grey)

The decision to select Southern Norway (NO2) instead of the originally proposed Northern Norway (NO4) was strategic. NO2 connects the Nordic hydro system to Continental Europe through major power cables to Germany and Denmark. Consequently, NO2 is exposed to cross-border price volatility that is not observed in the isolated northern zones, providing superior signals for forecasting. This selection methodology aligns with Mosquera-López et al. (2024), who prioritize data from representative geographical hubs when analyzing price dynamics.

4.1.1 Track A: Price Data Engineering

An Automated Ingestion Pipeline was developed to retrieve Day-Ahead prices from the ENTSO-E Transparency Platform. To handle the large 10-year dataset without causing errors, a Chunking Strategy was used. This method requests data in 6-month blocks and combines them, ensuring complete history from 2015 to 2025.

A technical challenge appeared in the Spanish market data (late 2025), where the data changed from hourly to 15-minute intervals. To prevent errors caused by mixed time formats, an automated resampling process was built into the pipeline. This logic detects 15-minute data and automatically averages it into hourly values. This ensures a consistent hourly format across the entire decade. A continuous timeline in UTC was created to align all zones. For missing data points, a Forward-Fill strategy was used. This implementation builds on the principles described by Aggarwal (2015), who notes the necessity of handling missing values using contextually nearby records in dependency-oriented data. In electricity markets, where prices often exhibit persistence between auctions, using the last valid observation serves as a practical application of this principle.

4.1.2 Track B: Weather Data and Spatial Aggregation

Unlike market data which comes from a single point, weather data comes in grids. The ERA5 Reanalysis dataset from the Copernicus Climate Data Store (CDS) is used for historical training data, which is widely recognized for its global accuracy. WeatherAPI.com is used for live data.

A key challenge was simplifying the complex ERA5 weather grids into a format the model could use. Instead of choosing a single location to represent a whole country, a Zonal Spatial Mean was implemented. This decision is supported by the findings of Neumann et al. (2023), who demonstrated that transforming grid-based weather data using spatial statistics (such as the mean) significantly improves forecast accuracy compared to using raw grid data. Following this logic, the map shapes of specific bidding zones are placed over the weather grid, and the average of all points inside the zone is calculated. Additionally, wind speed is calculated physically from its raw directional components—Zonal (u) and Meridional (v) velocity—to ensure accuracy ($Speed = \sqrt{(u^2 + v^2)}$), a calculation method also validated by Neumann et al. (2023).

For the live forecasting pipeline, data is collected from WeatherAPI.com. To match the training data, representative cities in each zone are queried and their results are averaged. Since the standard API often does not provide direct Solar Radiation data, a physics-based proxy was created. This estimates radiation based on the Ultraviolet (UV) Index ($Solar \approx UV * 25.0$). This ensures the model still sees accurate daily patterns even when direct sensors are unavailable.

4.1.3 Dataset Harmonization and Convergence

The final Mastersets are created by merging Track A and Track B on a unified UTC timeline. This step uses an "inner-join" method to ensure that only rows with complete price and weather data are kept for the model. Although the full ten-year history was collected, the data usage was split to address Concept Drift (structural changes in the market). The full history is used to find long-term seasonal patterns, but the forecasting model is trained mainly on data from after 2021. Recent literature highlights that the progressively increasing impact of renewable energy penetration fundamentally alters price dynamics. Limiting the training window prevents the model from learning old patterns that no longer apply, ensuring it adapts to changes caused by the 2022 energy crisis.

4.2 Pattern Detection

4.2.1 Proposed Solution vs Current Solution

The initially proposed pattern-detection workflow outlined a high-level sequence of steps: transforming hourly day-ahead prices into 24-dimensional daily vectors, standardizing the resulting profiles, applying Ward-linkage hierarchical clustering, determining an appropriate number of clusters, and interpreting the resulting day-types through centroid profiles and calendar-based visualizations. This workflow was intentionally designed as a structured, method-driven approach and largely follows established methodologies for clustering-based analysis of daily electricity price profiles as described in the literature, in particular the approach proposed by Roberts and Brown (2020).

As a first step prior to clustering, an exploratory data analysis was conducted across all three bidding zones (DK1, ES, NO2). The ten-year price histories clearly show substantial structural breaks: the early years exhibit very low volatility, followed by a sharp regime shift around 2020

with extreme price spikes and persistently higher volatility levels thereafter. Because these regimes differ fundamentally, clustering over the full ten-year horizon would artificially produce clusters that primarily separate “old stable years” from “recent volatile years,” rather than recurring daily price structures. To avoid mixing incomparable market phases, a timeline selection step was introduced, and the clustering analysis was restricted to the most recent three years, where all markets exhibit consistent and comparable dynamics.

This rationale is illustrated in figure 4, which shows the daily mean day-ahead price evolution for all three bidding zones over the full ten-year period. The x-axis represents time (years), and the y-axis shows daily mean prices in EUR/MWh. The figure highlights the clear structural break and sustained volatility increase, motivating the focus on a shorter and more homogeneous analysis window for clustering.

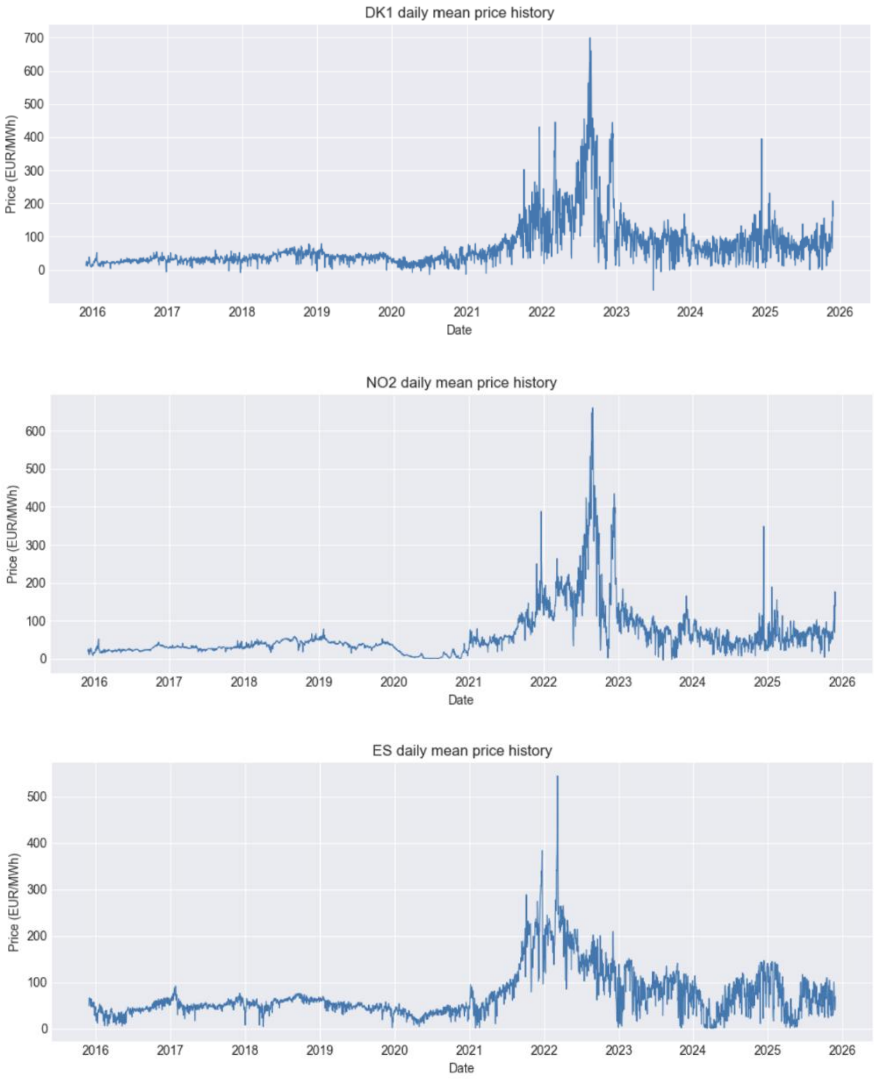

Figure 4 Historical evolution of daily mean electricity prices (EUR/MWh) for the selected bidding zones (DK1, NO2, ES)

Building on these exploratory findings, the clustering analysis was subsequently performed on the restricted three-year period. The implemented solution follows the proposed methodology almost exactly and consists of three core steps: daily vectorization of price profiles, hierarchical clustering using Ward linkage with Euclidean distance, and a systematic identification of an appropriate number of clusters.

Each day is represented as a 24-dimensional vector containing hourly day-ahead prices, allowing daily price shapes to be compared in a consistent and interpretable manner. Ward-linkage hierarchical clustering is then applied, which iteratively groups days minimizing the increase in within-cluster variance at each merge step, thereby favoring compact and internally consistent clusters. In practical terms, this means that days are clustered based on overall similarity in their hourly price profiles, resulting in groups that capture characteristic daily price regimes rather than isolated price events. This clustering strategy is well established in the literature for analyzing daily electricity price patterns (Roberts and Brown, 2020).

The only deliberate deviation from the original proposal concerns the vectorization step. Although standardization of daily price profiles was initially planned, the exploratory analysis showed that absolute price levels constitute a defining component of recent electricity price regimes. Standardizing the data would remove these level effects and obscure relevant structural information, particularly during the high-volatility period. Consequently, the final clustering approach is applied to raw daily price profiles, producing more realistic and interpretable regime patterns.

To determine a suitable number of clusters, several complementary diagnostics are jointly considered. First, an elbow-style plot of hierarchical merge distances is used, where the x-axis shows the number of clusters remaining after successive merge steps and the y-axis represents the merge distance at each clustering step. A visible flattening of the curve indicates diminishing gains from introducing additional clusters. Second, an acceleration plot is evaluated, where the x-axis corresponds to the clustering step and the y-axis shows the change in merge distances between successive steps. Pronounced changes in acceleration highlight cluster counts at which structurally distinct groups are merged. Finally, the distribution of observations across clusters is examined to avoid solutions dominated by very small or unbalanced clusters. Together, these criteria support the selection of cluster configurations that are both parsimonious and statistically robust.

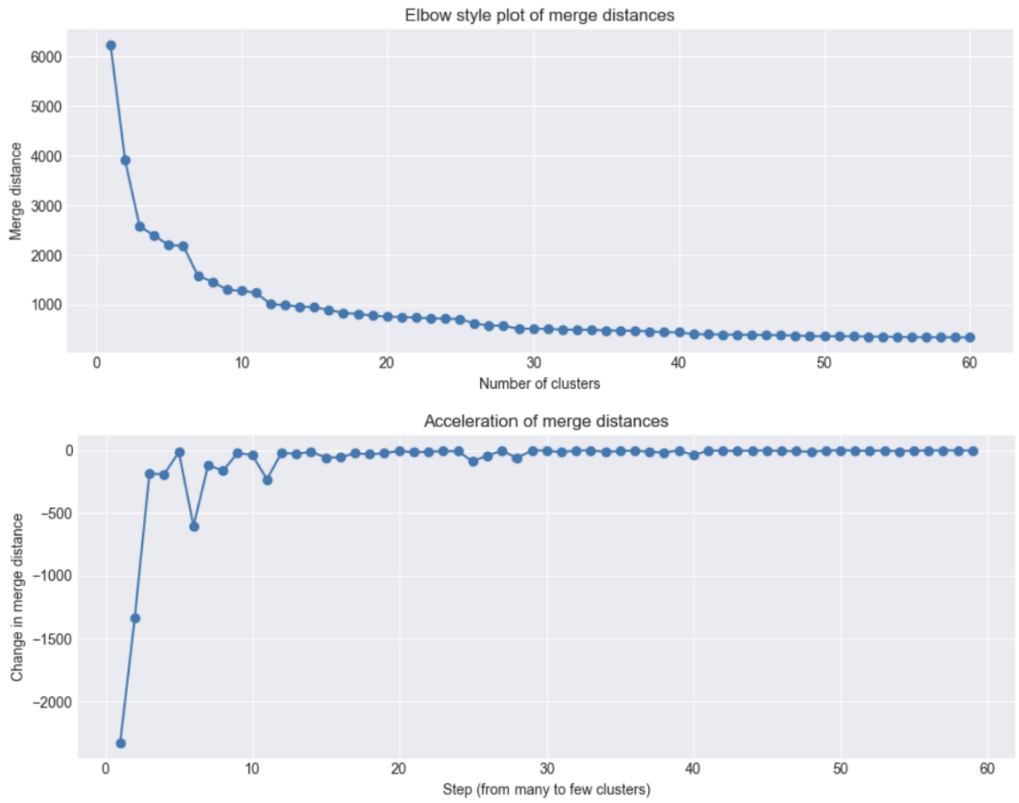

Figure 5 Determination of the optimal number of clusters using hierarchical clustering diagnostics. The top plot displays the Elbow Method applied to merge distances, where a flattening curve indicates diminishing gains from additional clusters. The bottom plot shows the acceleration of merge distances, highlighting structural breakpoints where distinct price regimes are merged.

For each selected cluster configuration, representative centroid profiles are computed. These centroids summarize the average hourly price shape of each cluster and form the basis for subsequent interpretation. To contextualize these profiles temporally, cluster assignments are mapped onto yearly calendar visualizations, enabling the identification of seasonal structures, weekday–weekend effects, and recurring price regimes over time. The resulting market intelligence insights derived from these visualizations are discussed in detail in Section 4.2.3.

Overall, the current solution remains closely aligned with the proposed architecture while incorporating insights from exploratory analysis and market-regime considerations, resulting in a clustering framework that is both methodologically sound and well suited to the structural characteristics of contemporary electricity markets.

4.2.2 Challenges

A central challenge in this work concerns the interpretability of the detected clusters. The clustering approach was initially motivated by the expectation that distinct daily price patterns would emerge, for example differences between seasons, weekdays and weekends, or recurring daily load structures. This expectation follows common findings in the literature, where clustering of daily electricity price profiles often reveals clearly distinguishable regime types (Roberts and Brown, 2020).

While the resulting clusters are statistically stable, many of them do not display sharply separated or easily distinguishable price profiles. Instead, the centroid shapes overlap to a considerable extent, which makes it difficult to describe individual clusters in a clear and intuitive way. In particular, assigning clusters to specific seasonal trends or calendar-related conditions proved challenging. As a consequence, the immediate interpretability of the clusters for market intelligence purposes is limited.

This difficulty is not purely methodological but also reflects recent market conditions. As discussed in the literature, periods of elevated volatility and structural market disruptions tend to blur traditional electricity price regimes, reducing the distinctness of typical daily price patterns (Roberts and Brown, 2020).

Several methodological adjustments were introduced to mitigate this challenge. The analysis was restricted from ten years to the most recent three-year period to avoid mixing fundamentally different market regimes. In addition, standardization of daily price profiles was omitted so that absolute price levels remain part of the clustering signal. Finally, different cluster configurations were evaluated to balance overly coarse groupings against excessively fragmented solutions. While these steps improved interpretability, a fundamental limitation remains: recent electricity price dynamics naturally produce less distinct and less intuitively separable cluster profiles.

4.2.3 Market Intelligence Insights

The clustering results identify recurring and interpretable price patterns across the three bidding zones over the observed three-year period. Instead of analysing individual daily price movements in isolation, the clustering approach groups days with similar intraday price behaviour into a small number of recurring price regimes. These regimes describe when certain price patterns occur, how persistent they are over time, and how they evolve across seasons.

From a market intelligence perspective, the clusters are useful in two main ways:

First, they provide insights relevant for market participation and trading. Some clusters are characterised by high intraday volatility and pronounced price peaks, while others exhibit flatter and more stable daily price profiles. High-volatility clusters indicate periods with larger intraday price spreads and therefore higher potential for short-term trading and arbitrage strategies. In contrast, low-volatility clusters correspond to calmer market conditions with fewer short-term opportunities. The clustering thus helps to distinguish phases in which active trading strategies are more attractive from phases dominated by stable price dynamics.

Second, the clusters offer a structured representation of price behaviour that can be used as an input for prediction models. Rather than modelling prices purely on a day-by-day basis, the clustering framework makes it possible to identify which type of price regime is currently present. Seasonal effects, weekday patterns, and regime persistence become visible at an aggregated level and can later be incorporated into forecasting models in a systematic and interpretable way.

Bidding Zone: DK1

The following analysis focuses on the bidding zone DK1 and illustrates how the identified clusters translate into recurring seasonal and weekly price regimes. The interpretation is based on three complementary elements: cluster centroids, cluster calendars, and the distribution of observations across clusters. Together, these representations describe how intraday prices typically evolve, when specific price regimes occur, and how frequently they appear.

The cluster centroids summarise typical intraday price profiles by showing the average hourly day-ahead price for each cluster. In the centroid plot, the x-axis represents the hour of the day (0–23), while the y-axis shows the average price level in EUR/MWh. Differences between

centroids therefore reflect both variations in overall price levels and differences in intraday price dynamics, such as morning and evening peaks.

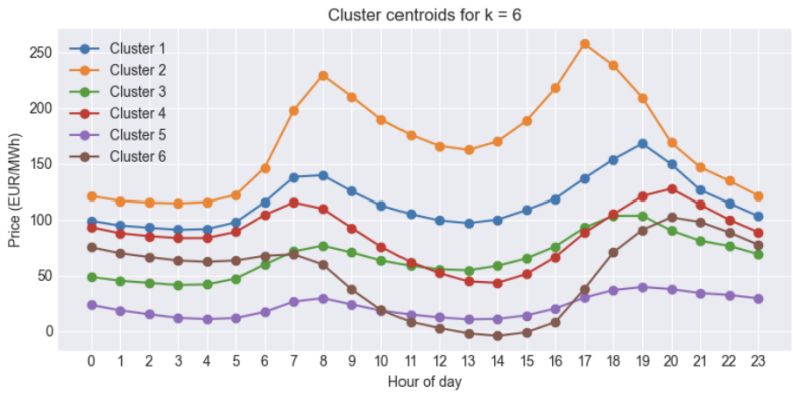

Figure 6 Cluster centroids for the DK1 bidding zone (k=6). The plots display the average intraday price profile for each of the six identified market regimes, illustrating distinct daily patterns in price levels (EUR/MWh) and volatility.

The cluster calendars complement this view by visualising when these price regimes occur over time. In the calendar plots, the x-axis represents weekdays (Monday to Sunday), while the y-axis shows the week index within the year. Each cell corresponds to one day and is coloured according to its assigned cluster, allowing seasonal patterns, weekday effects, and regime persistence to be identified at a glance.

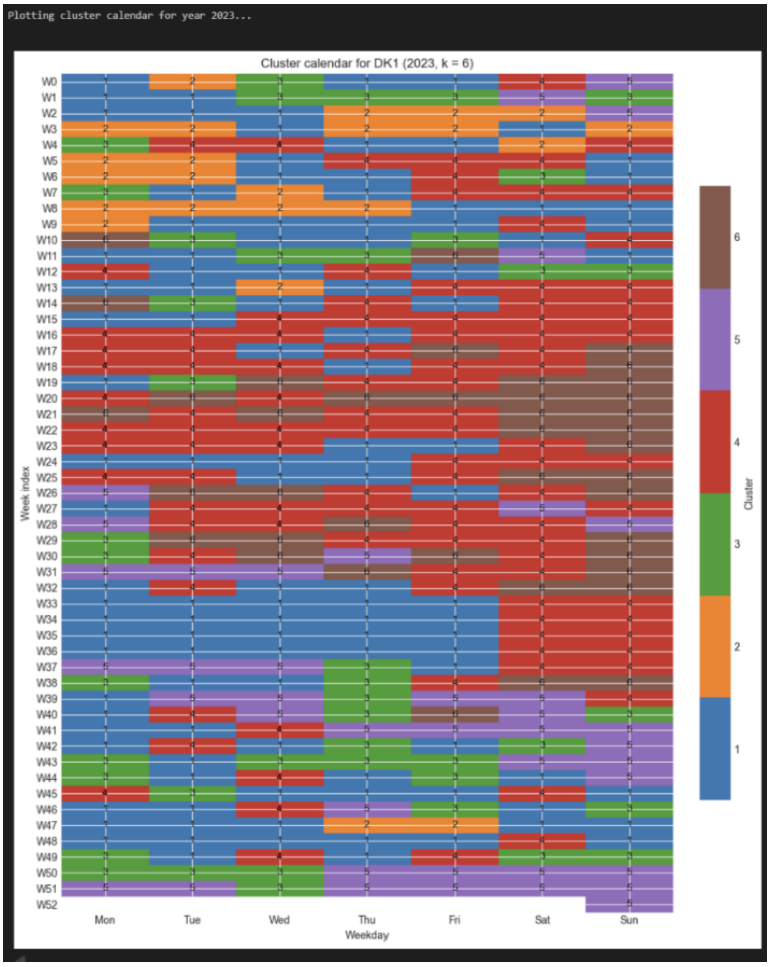

Figure 7 Cluster calendars for the DK1 bidding zone (2023). The heatmaps illustrate the temporal distribution of the six identified market regimes (k=6).

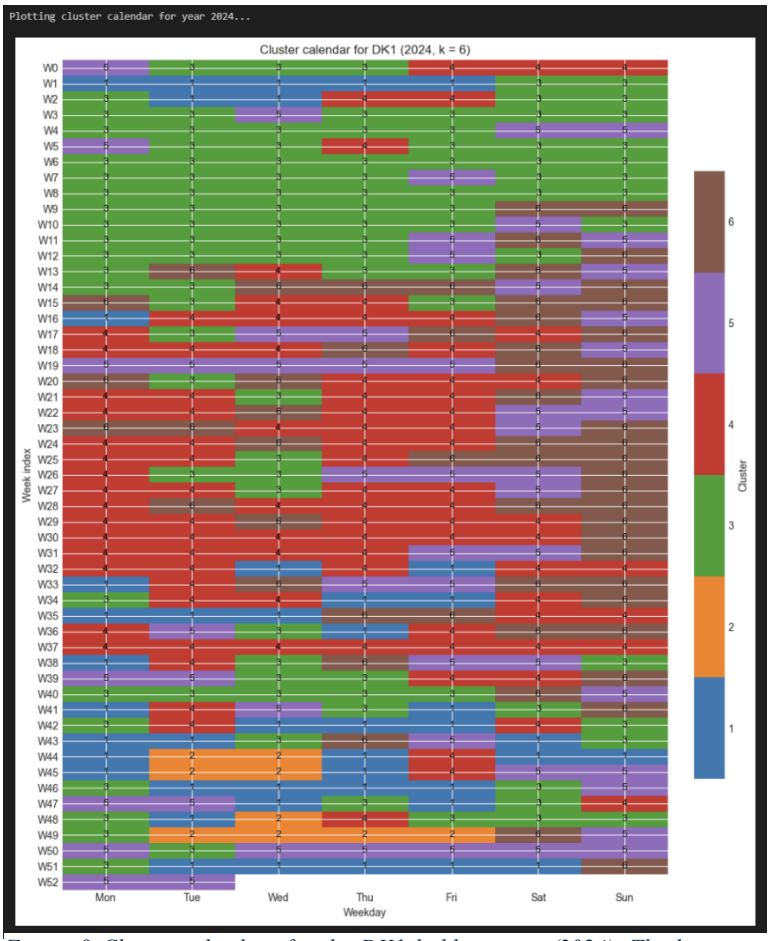

Figure 8 Cluster calendars for the DK1 bidding zone (2024). The heatmaps illustrate the temporal distribution of the six identified market regimes (k=6)

Figure 9 Cluster calendars for the DK1 bidding zone (2025). The heatmaps illustrate the temporal distribution of the six identified market regimes (k=6)

In addition, the cluster size distribution provides context on how frequently each regime occurs. The distribution shows that clusters are not equally sized, indicating that some price regimes are structurally more common in DK1, while others represent less frequent but potentially more distinctive market situations.

Cluster	Number of days
Cluster 1	222
Cluster 2	45
Cluster 3	187
Cluster 4	308
Cluster 5	119
Cluster 6	180

Table 1 Frequencies of the different clusters in DK1.

For DK1, the cluster analysis reveals a small number of recurring but clearly distinguishable patterns. Summer periods are most strongly associated with cluster 4 (red). The centroid profile of this cluster shows a medium overall price level combined with relatively high intraday volatility, indicating more dynamic price behaviour during summer months. This pattern is consistently visible in the cluster calendars. Seasonal patterns in winter are less pronounced. Winter days are most frequently assigned to cluster 1 (blue) and cluster 3 (green). Both clusters exhibit price levels similar to summer regimes but with noticeably lower intraday volatility, suggesting more stable daily price profiles rather than strong peaks. The clearest structure emerges for weekends during summer. These days are often dominated by cluster 6 (brown), which combines lower average price levels with high intraday volatility. This weekend effect is clearly visible in the calendars and reflects more irregular price dynamics on non-working days.

While these patterns are clearly observable, the clusters are not perfectly separable. Several centroid profiles partially overlap, and transitions between clusters are gradual rather than abrupt. As a result, the interpretation focuses on dominant tendencies rather than sharply distinct regime boundaries. Despite this limitation, the combined use of centroids, calendars, and cluster size information provides a coherent and interpretable description of recurring price regimes in DK1 and their seasonal and weekly structure.

Bidding Zone: NO2

The following analysis focuses on the bidding zone NO2 and illustrates how the identified clusters translate into recurring seasonal price regimes. As in the DK1 analysis, the interpretation is based on cluster centroids, cluster calendars, and the distribution of observations across clusters. A detailed explanation of the centroid and calendar visualisations is provided in the DK1 section and is not repeated here.

Figure 10 Cluster centroids for the NO2 bidding zone (k=5). The plots display the average intraday price profile for each of the five identified market regimes, illustrating distinct daily patterns in price levels (EUR/MWh) and volatility.

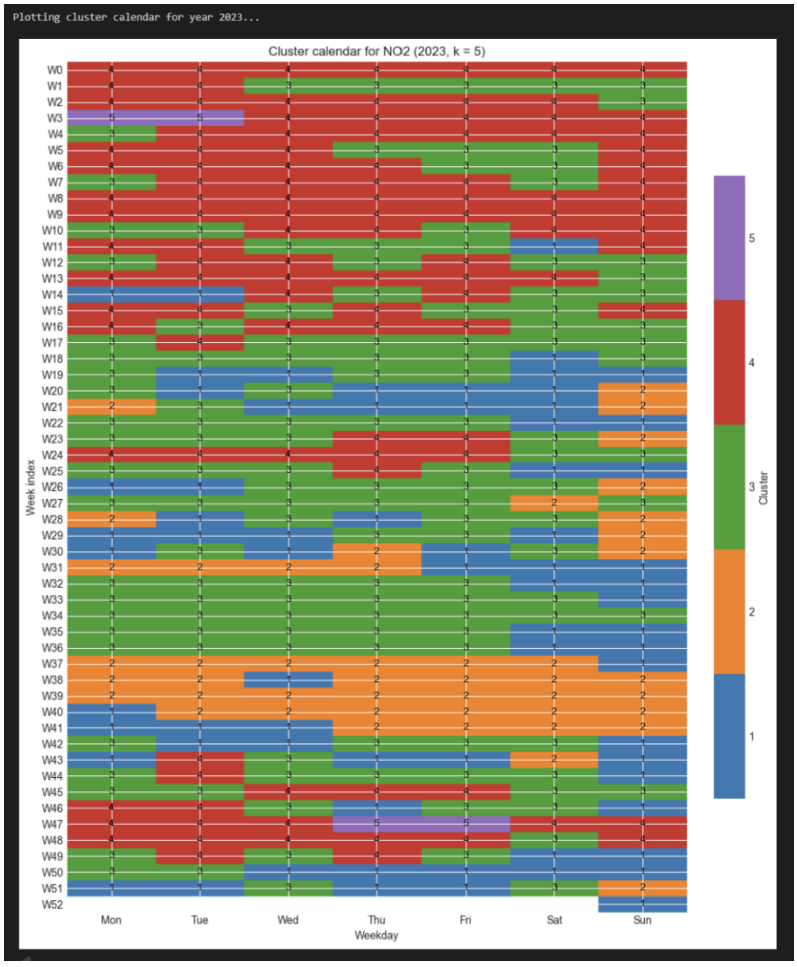

Figure 11 Cluster calendars for the NO2 bidding zone (2023). The heatmaps illustrate the temporal distribution of the five identified market regimes (k=5).

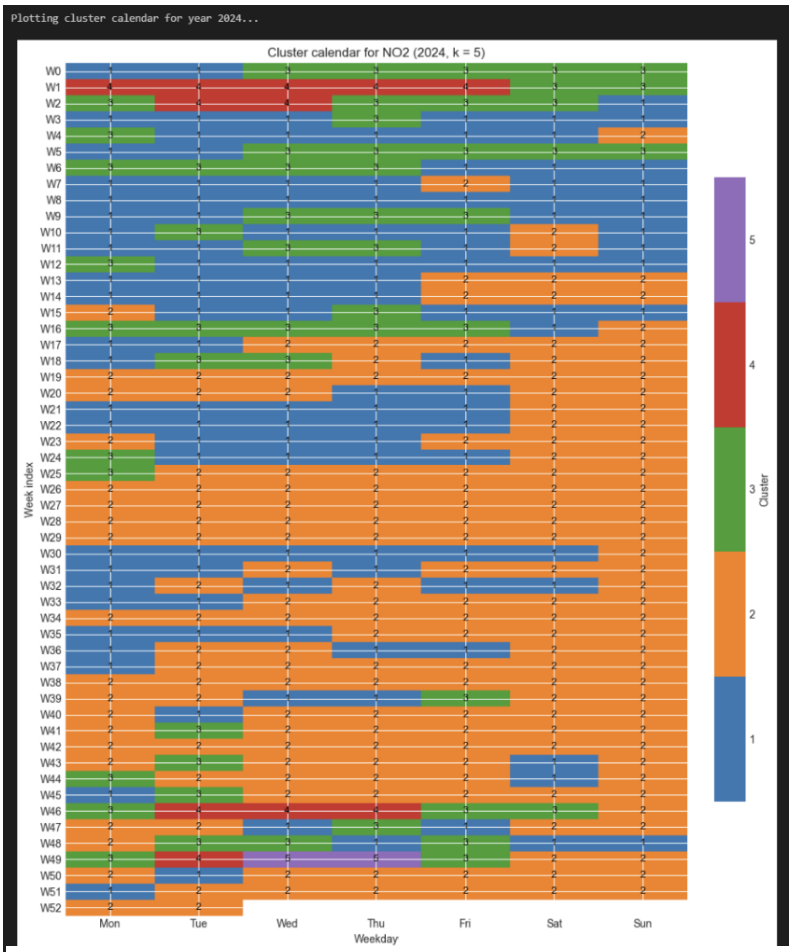

Figure 12 Cluster calendars for the NO2 bidding zone (2024). The heatmaps illustrate the temporal distribution of the five identified market regimes (k=5).

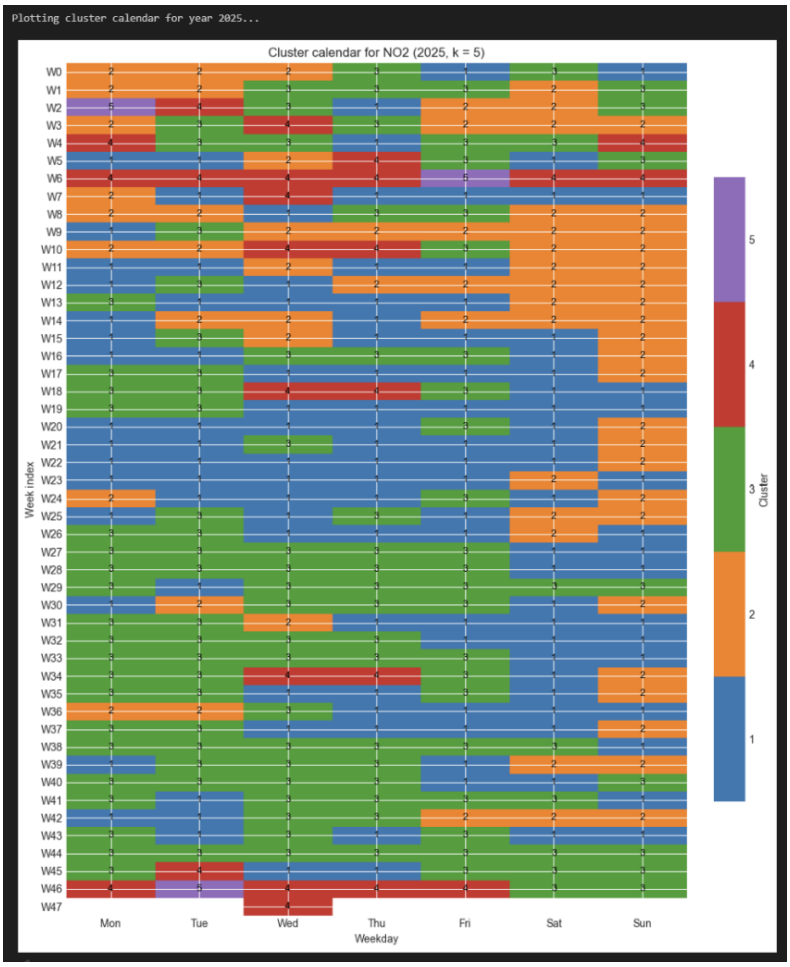

Figure 13 Cluster calendars for the NO2 bidding zone (2025). The heatmaps illustrate the temporal distribution of the five identified market regimes (k=5).

For NO2, the cluster calendars reveal a pronounced and recurring yearly cycle. Over the observed three-year period, the price dynamics typically alternate between phases dominated by cluster 1 (blue) and cluster 3 (green), followed by periods dominated by cluster 2 (orange) and cluster 4 (red), before returning again to cluster 1 and cluster 3. This sequence suggests a relatively stable and repeating seasonal structure in NO2.

Cluster 2 (orange) forms long and persistent stretches, most notably throughout 2024, indicating a stable and frequently occurring price regime. Cluster 4 (red) appears primarily during winter periods but decreases in frequency over the three-year window, suggesting a declining relevance of this regime in more recent years. Cluster 5 (purple) represents a rare high-price pattern and occurs only nine times in total, highlighting its exceptional nature rather than a recurring seasonal regime.

Cluster	Number of days
Cluster 1	312
Cluster 2	290
Cluster 3	306
Cluster 4	144
Cluster 5	9

Table 2 Frequencies of the different clusters in NO2.

The cluster size distribution confirms that price regimes in NO2 are not equally frequent. Some clusters represent dominant and persistent market conditions, while others capture infrequent but distinctive price situations.

As in DK1, the clusters are not perfectly separable. Several centroid profiles partially overlap, and transitions between regimes are gradual rather than abrupt. The interpretation therefore focuses on dominant seasonal tendencies and regime persistence rather than sharply defined boundaries. Overall, the combined evidence from centroids, calendars, and cluster frequencies provides a coherent description of recurring price regimes in NO2.

Bidding Zone: ES

The following analysis focuses on the bidding zone ES and illustrates how the identified clusters translate into recurring seasonal price regimes. As in the DK1 and NO2 analyses, the interpretation is based on cluster centroids, cluster calendars, and the distribution of observations across clusters. A detailed explanation of the centroid and calendar visualisations is provided in the DK1 section and is not repeated here.

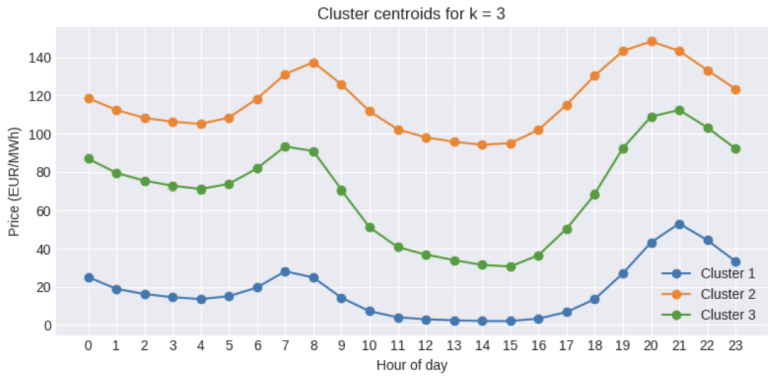

Figure 12 Cluster centroids for the ES bidding zone (k=3). The plots display the average intraday price profile for each of the three identified market regimes, illustrating distinct daily patterns in price levels (EUR/MWh) and volatility

Figure 13 Cluster calendars for the ES bidding zone (2023). The heatmaps illustrate the temporal distribution of the three identified market regimes (k=3).

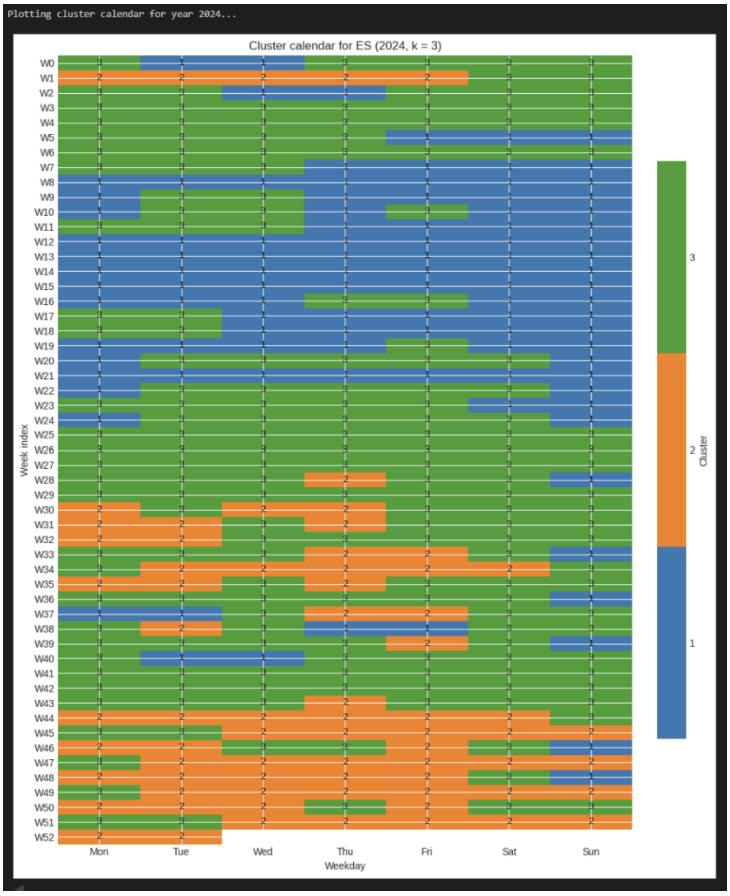

Figure 16 Cluster calendars for the ES bidding zone (2024). The heatmaps illustrate the temporal distribution of the three identified market regimes (k=3).

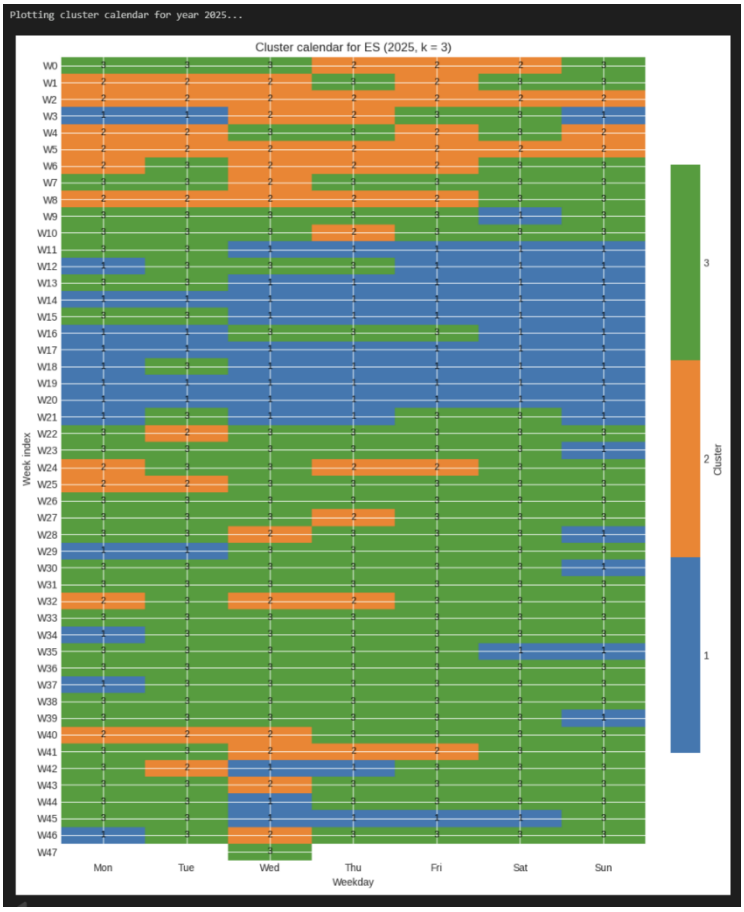

Figure 17 Cluster calendars for the ES bidding zone (2025). The heatmaps illustrate the temporal distribution of the three identified market regimes (k=3).

For ES, the cluster calendars indicate a clear but asymmetric seasonal structure. Cluster 1 (blue) is strongly associated with spring periods and appears consistently during these months across the observed years. In contrast, cluster 2 (orange) occurs more frequently during autumn and winter, pointing to higher price regimes in colder seasons.

Cluster 3 (green) does not follow a distinct seasonal pattern. Instead, it appears repeatedly in phases throughout the year and becomes increasingly dominant toward the end of the observation period. According to the centroid profiles, this cluster combines a medium overall price level with the highest intraday volatility, making it a particularly dynamic price regime. The recent increase in the prevalence of this cluster suggests a shift toward more volatile price behaviour in ES.

The cluster size distribution shows that the three clusters are relatively balanced in frequency, indicating that no single regime overwhelmingly dominates the ES market. This suggests a structurally diverse price environment in which multiple regimes coexist and alternate over time rather than a market driven by a single dominant pattern.

Cluster	Number of days
Cluster 1	226
Cluster 2	281
Cluster 3	554

Table 3 Frequencies of the different clusters in ES.

As in DK1 and NO2, the clusters are not perfectly separable. Several centroid profiles partially overlap, and transitions between regimes are gradual rather than abrupt. The interpretation therefore focuses on dominant tendencies and changes in regime prevalence rather than sharply defined boundaries. Overall, the combined evidence from centroids, calendars, and cluster frequencies provides a coherent description of recurring price regimes and their seasonal structure in ES.

4.3 Prediction Modelling

Predicting hourly day-ahead energy prices on the wholesale market forms the second part of this project, building on the recognized patterns and clusters from part one. Literature from the last years emphasizes the use of machine learning models for energy market predictions (Díaz et al., 2019; McHugh et al., 2020; Semmelmann et al., 2023). Some examples of used technologies include Gradient Boosting (Díaz et al., 2019; McHugh et al., 2020), Random Forests and XGBoost (McHugh et al., 2020; Semmelmann et al., 2023). Following the conclusions of McHugh et al. (2020), XGBoost is the most suitable of the three technologies. Therefore, for this project it was decided to go along with this insight and leverage XGBoost for prediction purposes.

For covariates a popular approach from the literature was followed by utilizing weather specific variables (Mosquera-López et al., 2024). Following the argumentation of Mosquera-López et al. (2024), suitable weather variables are temperature, wind speed, irradiance and precipitation. Even though precipitation has delayed effects on energy generation, since rainwater is first stored in reservoirs before it is used to generate electricity, Mosquera-López et al. (2024) argue that it is a more suitable covariate than reservoir levels. While the study of Mosquera-López et al. (2024) also highlights the strong influence of fundamental drivers, such as natural gas prices, this project only focused on the predictive power of weather features, as it was agreed with the project partner that other variables than weather related ones, for explanatory purposes, are out of scope.

4.3.1 Methodology

Building on the results from the first part of the project, to obtain final hourly price predictions, the prediction model of each bidding zone consists of two layers and follows three steps as described below.

4.3.1.1 Cluster predictions (Layer 1)

After insights into the occurrence of certain price clusters within each bidding zone considered were generated in the first part of the project, these insights are leveraged in step one of the prediction modelling. The cluster analysis delivered a calendar-like classification of every day, in the 3 years under exploration, into a certain cluster. These classifications were utilized as

target variables, to create one XGBoost classification model for each bidding zone, with the ability to predict the probabilities of the different clusters possible within a bidding zone to occur on a certain day. For example, as in the Denmark zone 6 clusters were recognized, the output of the XGBoost classification model for the Denmark zone are probabilities, representing the likelihood of occurrence for each of the 6 clusters on a certain day. The covariates for these models are

- Mean temperature,
- Mean wind speed,
- Sum of precipitation,
- Sum of irradiance,
- Calendarial dummy indicating if a day is part of the weekend or not,
- 3 calendarial dummies indicating the season a day is part of (spring, summer, autumn, winter)
- Mean energy price from one day prior,
- Mean energy price from two days prior,
- Mean energy price from three days prior.

4.3.1.2 Within cluster price predictions (Layer 2)

With step 2, XGBoost regression models with the hourly energy price as target variable are trained. Such models are trained for each cluster found in part one of the project. Therefore, the data from all days falling within a certain cluster is gathered and considered independently from the other days. The explanatory variables for these models are

- Temperature at the respective hour,
- Wind speed at the respective hour,
- Irradiance at the respective hour,
- Aggregated precipitation from the last x days, with $x \in \{3, 7, 14, 20\}$.

Since literature suggests delayed effects of up to 20 days for precipitation on energy prices (Masquera-López et al., 2024), to find the optimal x, each cluster was evaluated with the values 3, 7, 14 and 20 for x. With 6 clusters identified in the Denmark zone, 3 clusters in the Spain zone and 5 clusters in the Norway zone, each cluster trained with 4 different values for x, a total of 56 models, each predicting the hourly energy price within a cluster, were obtained. Results suggest an optimal x of 20 for Denmark and Spain, while for Norway it is 7. Therefore, only

the regression models leveraging these numbers of days for aggregated precipitation in the respective bidding zones are utilized in the last step.

4.3.1.3 Obtaining final price predictions

To obtain one final price prediction per hour and bidding zone the models from layer 1 and layer 2 are combined. Leveraging the models from layer 1, for each cluster the likelihood of it appearing is calculated. Next, the models from layer 2 are utilized to calculate a price prediction for each cluster, based on the above defined weather variables. Lastly, the predictions from each cluster are weighted with the appearance likelihood of the respective cluster, resulting in one final value per hour and bidding zone.

4.3.2 Evaluation

The chosen metric for the evaluation of the models is the Weighted Absolute Percentage Error (WAPE). Since it provides a robust, stable and volume-weighted measure for forecast quality, directly reflecting business impact for energy trading companies (Chockalingam, 2023), it serves as a suitable metric for evaluating energy price predictions. Furthermore, traditional metrics, like MAPE (Mean Absolute Percentage Error) can be very unstable, as they are tendentially easier skewed by small actual values (Chockalingam, 2023). Therefore, the WAPE has already been in use as evaluation metric and benchmark criterium in Malonzo et al. (2025).

With y_t representing the real value at time t, $\hat{y}_t$ the predicted value at time t and n the number of observations, the WAPE is defined as

$$WAPE = \frac{\sum_{t=1}^{n} |y_t - \hat{y}_t|}{\sum_{t=1}^{n} |y_t|} * 100\%.$$

4.3.3 Benchmarks

In order to create benchmarking possibilities for the modelling approach of this project, at the beginning of December it was mutually agreed with the project partner to create two different naïve approaches for energy price predictions.

4.3.3.1 Naïve approach one

For naïve approach one, the hourly energy prices of the previous day are utilized as predictions for the following day. This means e.g. the price prediction of November 2nd at 2pm equals the real price from November 1st at 2pm.

4.3.3.2 Naïve approach two

Naïve approach two directly leverages the weather variables utilized in layer 2 of the modelling approach defined above, meaning the hourly energy prices are predicted using only

- Temperature at the respective hour,
- Wind speed at the respective hour,
- Irradiance at the respective hour,
- Aggregated precipitation from the last x days (x = 20 for Denmark zone and Spain zone; x = 7 for Norway zone), without using clusters at all.

4.3.4 Results

The results indicate the modelling approach tested in this project to deliver predictions worthy of deeper exploration, but not good enough quality to already deliver trading ready suggestions. Compared to naïve approach 2, the cluster-based models of this project performed better in all bidding zones. In Denmark, these cluster-based models even performed best out of all model variations with a WAPE score of 31.45%. For Spain and Norway, the cluster-based approach came in second, after naïve approach 1, lacking behind only 0.82% in Norway and 4,24% in Norway. Table 4 and figure 18 display the results of all three approaches in the different bidding zones.

	Bidding zone	WAPE [%]
Cluster-based models	Denmark	31.45
	Spain	33.84
	Norway	23.28
Naïve approach 1	Denmark	34.39
	Spain	29.60
	Norway	22.46
Naïve approach 2	Denmark	33.56
	Spain	40.70
	Norway	30.49

Table 4 A table representing the results of the different modelling approaches in each bidding zone, measured with the WAPE.

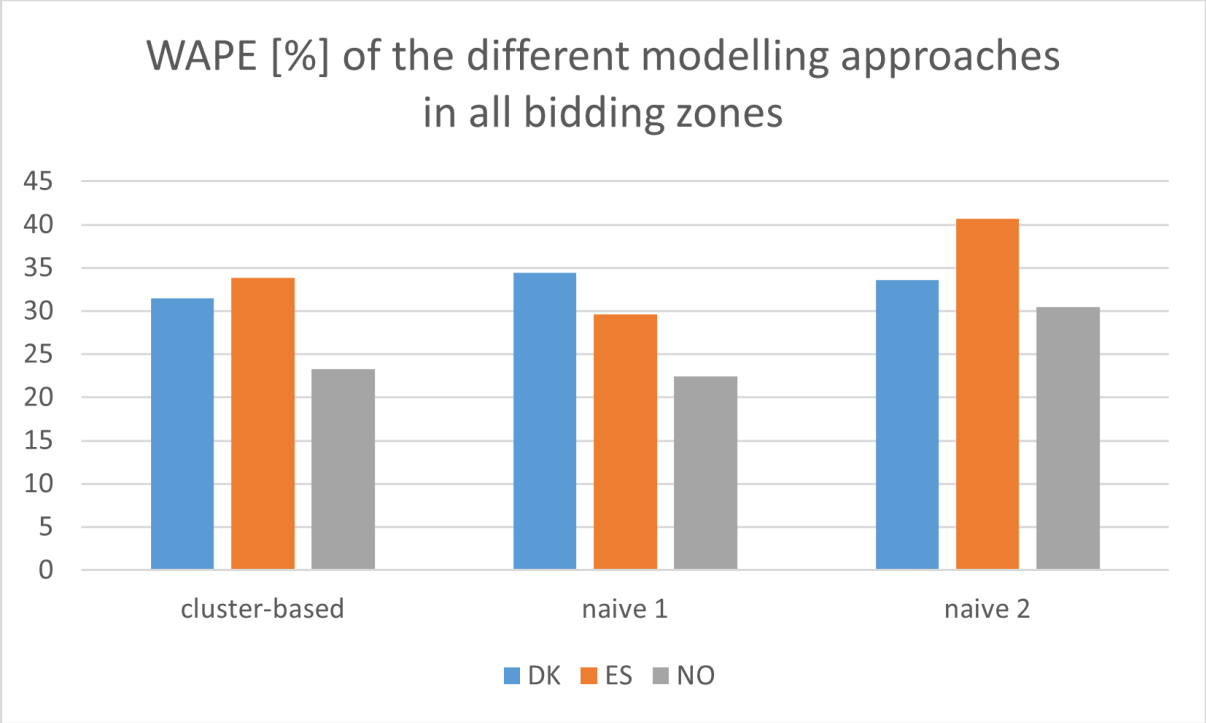

Figure 14: A figure visualizing the results of the different modelling approaches, measured with the WAPE.

To give a better understanding, figure 19 visualizes the predicted versus the actual price over the course of 48 hours for the three different modelling approaches. These 48 hours span over June 21st, 22nd and 23rd 2025 and represent the Denmark zone.

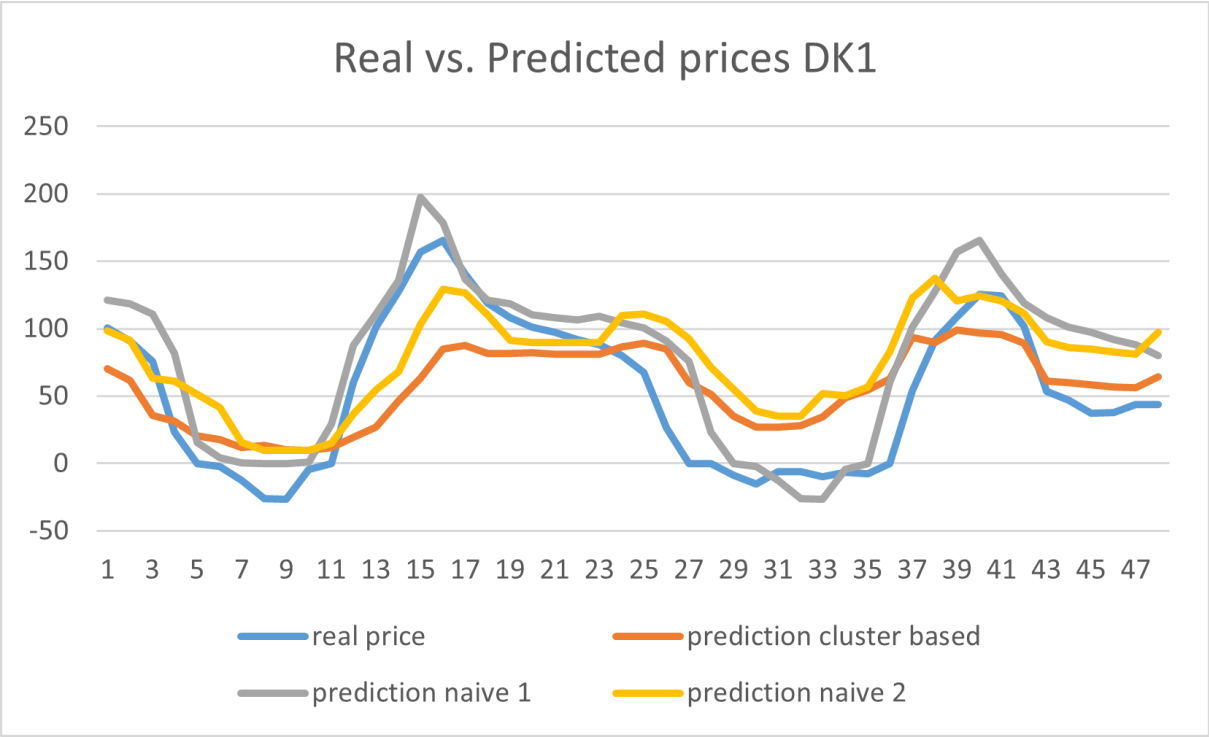

Figure 15: A line graph visualizing the real vs. the predicted prices over the course of 48 hours in the Denmark zone.

4.4 The MARBL Energy Dashboard

To translate data engineering and machine learning models into actionable business insights, a production-grade application was developed. The MARBL Dashboard serves as the unified interface for historical market data, unsupervised learning patterns, and real-time machine learning forecasts. The application was built using the Streamlit framework, chosen for its ability to rapidly deploy interactive data visualizations while maintaining high performance.

The system architecture follows a modular Model-View-Controller (MVC) pattern, decoupling the inference engine from the user interface to ensure stability. While the machine learning models were trained on high-precision historical reanalysis data (ERA5) to ensure robust pattern detection, the live inference engine requires real-time data availability. Therefore, the dashboard utilizes the WeatherAPI service to fetch Day-Ahead weather forecasts (t + 24h), which are then processed through the same feature engineering pipeline (e.g., wind magnitude calculations) to ensure consistency with the training features. To prevent downtime during API outages, a Fallback Simulation Mode was implemented, which automatically switches to synthetic data to maintain dashboard functionality.

4.4.1 Application Architecture

The system follows a modular software design pattern that separates the visual presentation from the underlying calculations. This approach ensures that the application remains maintainable and scalable. The architecture is divided into three distinct layers where the entry point acts as the central controller. The presentation layer consists of dedicated pages for specific analytical workflows while the utility layer handles all background tasks including data access and machine learning inference. To ensure a responsive user experience, advanced performance optimization techniques were implemented. Heavy data operations are managed through a caching system that keeps frequently accessed data in memory which allows for sub-second page reloads. Additionally, the system employs defensive programming strategies so that if live data sources become unavailable, the application automatically switches to a fallback simulation mode to prevent crashes.

4.4.2 User Interface and Analytical Workflow

The dashboard is organized into three specific analytical modules that are designed to answer different business questions. The first module focuses on the analysis of historical trends where dynamic filters allow the user to slice data by specific bidding zones and date ranges. A key feature is the profile analysis which calculates and visualizes the average daily price curve along with standard deviation bands. This visualizes the intraday volatility and allows users to quickly identify standard market behaviors such as the solar-driven price dip in Spain or the morning consumption peak in Norway.

The second module visualizes the results of the unsupervised learning algorithms. Instead of showing raw data, this page displays the representative shapes of the identified market regimes. For example, specific clusters representing high wind days or summer solar peaks are mapped to descriptions that are easy to understand. A custom calendar heatmap is used to show how these market regimes evolve over weeks and months which makes abstract machine learning results interpretable for human analysts.

The final module provides day-ahead price predictions. To reflect the uncertainty inherent in energy markets, the display is probabilistic. It shows not only the predicted price but also the confidence the model has in specific market regimes. Transparency is prioritized by explicitly

displaying the weather forecast inputs such as wind speed and cloud cover that are driving the current prediction.

4.4.3 Inference Engine and Backend Logic

The core intelligence of the dashboard is powered by a combined model strategy. This engine moves beyond simple regression by splitting the prediction task into two stages. First, a classification model predicts the probability of the current day belonging to a specific market cluster. Second, specialized regression models predict the hourly prices. The final output is a weighted average of these specialized predictions. This backend logic also handles complex feature engineering where raw weather data fetched from the ingestion pipeline is transformed into model-ready features. This includes calculating rolling averages for prices to capture trends and aggregating precipitation data over varying windows to accurately model the differing hydro-reservoir dynamics of each region.

4.4.4 Robustness and Future Work

The design of the data and dashboard pipeline prioritizes robustness and adaptability which makes it highly suitable for future expansions. Since the ingestion logic is decoupled from the visualization layer, adding new bidding zones or integrating alternative pricing models can be done with minimal code changes. The underlying data structure was designed to be agnostic to the specific region so incorporating a new country would only require adding the relevant API keys and shapefiles. This modular foundation ensures that the system can easily evolve to support larger datasets or more complex forecasting algorithms in future iterations of the project.

5 Recommendations for a possible follow-up project

In case this project is carried on in the future, the team suggests to carry out these next steps.

- Find out when exactly the cluster-based models delivered good predictions (What were the circumstances at these times?).
- Analyze, why the cluster-based models performed comparatively well in Denmark.
- Consider additional explanatory variables, not only weather-based variables.
- Test different models than XGBoost (eventually Neural Networks).
- Expand Geographic Scope: Leverage the modular ingestion pipeline to include interconnected bidding zones like Germany (DE-LU) or the Netherlands (NL). Since the automated chunking and resampling logic is already zone-agnostic, this would require minimal code changes and would capture critical cross-border price influences.
- Enhance Dashboard Interactivity: Extend the Streamlit dashboard to allow "What-If" analysis. Currently, the dashboard shows a static forecast; future work could allow users to manually adjust weather inputs (e.g., "What if wind speed drops by 50%?") to see the immediate impact on predicted electricity prices.

6 Team Roles and Responsibilities

The project team consists of four Business and Data Science students. While responsibilities were clearly distributed, all members collaborated closely across the entire workflow, contributing to analysis, modelling, feedback, and documentation to ensure a coherent and high-quality project outcome.

Maximilian Dieringer

Maximilian serves as the internal Project Coordinator, overseeing team organization, timeline management, and the preparation of presentations and reporting. His primary technical responsibility lies in the complete clustering and pattern detection component, including methodology, implementation, interpretation, and visualization. In addition, he contributed to research on relevant data sources and scientific literature, informing both the proposal and the methodological setup. While Maxim carried out the data integration and preprocessing, Maximilian supported this process as a sparring partner by reviewing pipeline logic, providing feedback, and addressing conceptual questions. He furthermore supported the dashboard development as a sparring partner through reviews and feedback and conducted research on suitable visualization tools, which was essential for the final tool selection.

Maxim Gomez Valverde

Maxim is responsible for the full data integration and preprocessing workflow. He implemented the API calls, managed the extraction of large-scale historical datasets, and resolved the technical challenges associated with collecting and harmonizing the raw data. He also set up and maintained the GitHub repository, ensuring reproducibility, structure, and version control. In addition, Maxim was technically responsible for building the dashboard, including its implementation and integration with the processed data.

Harald Körbel

Harald is the primary contact for external communication with MARBL and coordinates all interaction with the project partner. Together with Daniel, he developed the conceptual forecasting model design and contributed to literature research related to modeling approaches and covariates. He also supported the search and evaluation of data sources relevant for the forecasting stage. His primary technical responsibility lies in the forecasting component, including the full implementation and evaluation of all forecasting models.

Daniel Klaric

Daniel served initially as the main contact with the academic supervisor. He conducted research on data sources and forecasting-related literature and developed the forecasting model concept together with Harald. While Maxim executed the preprocessing pipeline, Daniel acted as an active sparring partner by providing feedback, supporting debugging discussions, and helping address arising challenges.

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